

### Textbook & Materials:

- There is no textbook required for this course. Occasional readings and videos will be posted to Canvas.
- Access to Solidworks. Solidworks is available in the engineering lab and can be accessed online. Contact [REDACTED]  
[REDACTED]
- Lab tools and equipment access. All students registered for the class will have keycard access to the engineering lab (SCI 014). Students must be trained on safety and operation before use of any equipment or tool.

### Course Description and Prerequisites:

Provides the student engineer with firsthand experience in moving from a stated need to a developed and proof-tested product. Topics include project logbooks and plans, evaluating concepts and selecting a design, preparing design documents, fabrication, development and testing of prototypes, stewardship of the environment, preparation of engineering reports, and principles of contract, engineering, and patent law. Prerequisites: ENGR 201.

### Course Objectives:

As engineers, we are being equipped with scientific training, technical skills, and engineering capabilities that can improve the lives of others and the world around us. However, without the ability to listen contextually and empathize, recognize and define problems, communicate effectively, and collaborate with others from diverse backgrounds, we limit our potential impact for solving real world issues and problems. In this course, we will be exploring how actively engaging individuals and communities, is a more effective way to design, and is consistent with a Biblical view of justice, service, humility, and love.

After satisfactory participation in this course, students should expect to be able to identify with the following course-level learning outcomes.

### *Course-Level Learning Outcomes:*

- A. I will be able to apply the design process to serve others and Christ's kingdom and deliver solutions to real engineering problems using a human-centered design approach (empathize, define, ideate, prototype, test).
- B. I will be able to gather and define user needs, write requirements and specifications, and perform verification of designs.
- C. I will be able to document and communicate results effectively using proper technical language and prose via engineering reports, various presentation formats, and product documentation.
- D. I will be able to apply prototyping techniques at varying levels of fidelity to communicate and demonstrate a specific design.
- E. I will be able to work collaboratively in a diverse design team to advance an engineering project within a structured timeline.
- F. I will better understand my own cultural identity, values, motivations, and biases and how this dictates my position in relationship to a cross-cultural team.

## Grading:

This is a collaborative, project-based course. Grading will depend on performance for both individual and team-based deliverables. Specific grading criteria and rubrics will be provided for each deliverable in class. See schedule below for deliverable assign and due dates.

### ***Individual Deliverables: 35%***

#### *Design Logbook: 15%*

- *Logbook Expectations:*  
Each student will be required to keep an updated design logbook. Your logbook should represent the current status of your projects. Logbooks will be reviewed for grades three times through the semester. Logbook grades will be based on quality of logbook content and how up to date the logbook is.
- *Purchasing a Logbook:*  
The design logbook can be an 8.5" x 11" spiral bound notebook, or hardcover standard bound notebook.
  - Moleskine Art Plus Sketchbook (5" x 8.25")
  - TOPS Docket Quadrille Pad, Wire Bound, 8-1/2 x 11-3/4 Inches, Quad Rule (4 x 4)
- *Logbook Entries:*  
This notebook should be used for documenting sketches, concept ideas and details of the overall design process. Team meeting minutes as well as team assignments and action items should be recorded in logbooks. This can be considered a legal document and is an important part of the design process. Each entry should have a title, page number and date. All sketches should be annotated so that anyone can interpret the sketch.

#### *Personal Design Project: 15%*

- Each student will complete a personal design project during the semester. We will apply the design process to ourselves as we work to find and solve a personal problem. Credit is based on successfully completing the project based on your own written objectives and appropriate application of the design process modes and methods.

#### *Design Challenge Presentation: 5%*

- Each student will be required to deliver a Design Challenge presentation in class.
- Each presentation will be less than 5 minutes.
- Each presentation will be accompanied by a single concept slide. Details will be given in class.

### ***Team Project Deliverables: 65%***

Each team will be responsible for developing a novel product or service from concept to functional prototype that solves a real-world, stated problem. Students will utilize the design thinking process and will complete the following team-based deliverables.

#### *Team Charter: 5%*

- Each team will develop a charter that defines how their team will operate. Charters should include a code of ethics, team expectations, specific plans and methods for meeting in and outside of class for project work, team roles, and a conflict resolution method.

#### *Sketch Model Review: 10%*

- Each team will create detailed sketches and build concept prototypes with craft materials that communicate the feel and function of their concept back to the user.

- Each team will submit a single page executive summary outlining their work on the Sketch Model, latest project findings, and the planned project path forward.

#### *Assembly Model Review: 10%*

- Each team will present a detailed assembly model of their product design utilizing Solidworks or similar design software. The purpose of the Assembly Model Review is to create team consensus of product architecture and create a forum for design feedback.
- Each team will submit a single page executive summary outlining their work on the Assembly Model, latest project findings, and the planned project path forward.

#### *Feasibility Prototype: 10%*

- Each team will build an initial prototype that demonstrates the proper design capability of the product. Initial product testing should be performed with the feasibility prototype.
- Each team will submit a single page executive summary outlining their work on the Feasibility Prototype, latest project findings, and the planned project path forward.

#### *Functional Prototype: 10%*

- Each team will build a functional prototype that demonstrates the proper form and function of the product.

#### *Final Presentation: 10%*

- The final technical presentation will be submitted including functional prototype, CAD drawings, testing results and conclusions. This presentation should include an initial product offering.

#### *Final Report: 10%*

- The final technical report will include a narrative of the development project, key project discoveries, CAD drawings, testing results, planned next steps, and conclusions.

#### *Self- and Peer-Assessment*

- A mid-term self- and peer-assessment will be conducted during the semester. This review will allow you to receive feedback on your performance within your team. Team grades may be adjusted for individual scores based on assessments.

#### **Grading Scale:**

Individual grades will be given according to following grading scale:

| <i>Letter Grade</i> | <i>Lower Cutoff</i> | <i>Letter Grade</i> | <i>Lower Cutoff</i> |
|---------------------|---------------------|---------------------|---------------------|
| A                   | 93%                 | C+                  | 77%                 |
| A-                  | 90%                 | C                   | 73%                 |
| B+                  | 87%                 | C-                  | 70%                 |
| B                   | 83%                 | D                   | 60%                 |
| B-                  | 80%                 | F                   | 0%                  |

#### **Class Periods:**

This is a collaborative, project-based course. It is important that you are present and fully engaged during class periods. Each absence without valid excuse and prior permission from the instructor and your teammates will result in a 1% final grade decrease. Leaving before the class session is over without permission or being excessively late will also count as an

absence. Part of class time will be regularly devoted to project work. Always bring your laptops, logbooks, and lab-safe attire.

### **Workshops:**

Several workshops from department professors, lab TA's, or guest speakers will be scheduled either during or outside of class time during the semester. These workshops will cover topics as needed for project work such as Presentation Skills, Technical Writing, Solidworks Modeling, Fabrication and Prototyping, Electronics, Basic Tools Training, Heavy Equipment Training, Project Scheduling and others. Workshop session schedule and sign-ups will be announced in class.

### **Late Assignment Policy:**

Individual assignments and project deliverables must be completed and submitted by the due time and date. Late assignments will not be accepted without first meeting with the instructor and receiving an accepted excuse.

### **Class Communication:**

Course assignments, announcements and documents will be communicated in class or through Schoology. Questions or comments regarding assignments, projects, or class material are welcome through the Schoology site or by e-mail. Grades will be posted regularly through Schoology.

### **Accommodations:**

Wheaton College is committed to providing access and inclusion for all persons with disabilities, inside and outside the classroom. Students are encouraged to discuss with their professors if they foresee any disability-related barriers in a course. Students who need accommodations in order to fully access this course's content or any part of the learning experience should connect with Learning and Accessibility Services (LAS) as soon as possible to request accommodations <http://wheaton.edu/las> (Student Services Building - Suite 209, [REDACTED] phone [REDACTED])

hesitate to reach out or ask any questions.

### **Academic Honesty:**

Collaboration with your classmates on team-based deliverables is a necessity. However, each team member is expected to contribute equally to the project. This means dividing up tasks appropriately, contributing to team meetings, discussions, and reports, and attending all meetings and classes on time. Be respectful, responsible, proactive, and ethical as a dedicated and committed team member. Any assignments which demonstrate plagiarism from classmates or external sources will receive no credit. See the Wheaton College [Integrity of Scholarship](#) Policy for details.

### **Diversity & Respect:**

Please be aware of Wheaton College's policy on inclusive language, "For academic discourse, spoken and written, the faculty expects students to use gender inclusive language for human beings."

### **Confidentiality and Mandatory Reporting:**

As an instructor, one of my responsibilities is to help create a safe learning environment on our campus. I also have a mandatory reporting responsibility related to my role as a faculty member. I am required to share information regarding sexual misconduct about a crime that may have occurred on Wheaton College's campus with the College. Confidential resources available to students include Confidential Advisors, the Counseling Center, Student Health Services, and the Chaplain's Office. More information on these resources and College Policies is available at [www.wheaton.edu/sexualassaultresponse](http://www.wheaton.edu/sexualassaultresponse)

**Writing Center:**

The Writing Center is a free resource that equips undergraduate and graduate students across the disciplines to develop effective writing skills and processes. This academic year, the Writing Center is offering in-person consultations in our Center in Buswell Library, as well as synchronous video consultations online. Make a one-on-one appointment with a writing consultant [here](#).